Unit 3.1 Teacher Key: Age of the Earth

Answers for every printable, plus the answer notes from each lesson. Student versions are in Unit31_Printables.pdf.

Day	Printable	Use	Copies
Day 1	Unit CER page: question, claim, evidence log	Glue in · write on it	1 per student
Day 1	Frayer card sheet (homework)	Glue in · write on it	1 per student
Day 1	Hook data: ocean floor and continent ages	Glue in · reference	1 per student
Day 1	Site list for the plotted map	Glue in · write on it	1 per pair
Day 1	Plate boundary table	Glue in · write on it	1 per student
Day 2	Seafloor ages and color key	Glue in · reference	1 per student
Day 2	Seafloor strip	Glue in · write on it	1 per student
Day 2	Graph grid: age vs. distance	Glue in · write on it	1 per student
Day 2	North America crust ages and strip	Glue in · reference and write on it	1 per student
Day 3	Crater counts	Glue in · write on it	1 per student
Day 3	Timeline table and old materials	Glue in · reference and write on it	1 per student
Day 4	Guided notes	Glue in · write on it	1 per student
Day 4	Density table	Glue in · write on it	1 per student
Day 4	SketchNotes checklist	Glue in · write on it	1 per student
Day 5	Magnetic stripe bands	Glue in · reference	1 per student
Day 5	Hawaii and GPS	Glue in · reference and write on it	1 per student
Day 5	Fossil table	Glue in · reference	1 per student
Day 5	Evaluate the evidence	Glue in · write on it	1 per student
Day 6	Three buckets	Glue in · write on it	1 per student

Day 6	Part A sentence frames	Glue in · reference	1 per student
Day 6	Peer feedback	Glue in · write on it	1 per student
Day 6	Evidence cards	Glue in · write on it	1 per student
Day 6	Part B story frames	Glue in · write on it	1 per student
Day 7	Self-rating chart	Glue in · write on it	1 per student

Day 8 (CSA) needs nothing glued in. Not printed here: the world map with latitude and longitude lines (Day 1), the continent cut-outs for the Pangaea puzzle (Day 5), and the 45 cm adding-machine tape strips (Day 3).

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Unit CER page: question, claim, evidence log

Your first claim goes under the question. Add a row each day. Row 1 is done for you. Glue this on its own new page.

Question: Why is the seafloor so much younger than the continents?

My first claim (Day 1):

The seafloor is young because it keeps getting renewed. (Baseline: accept any attempt.)

Day	Evidence (use a number)	What it shows
1	The oldest ocean floor is about 200 million years old. Continental rock is up to 4.4 billion years old.	The seafloor is much younger than the continents.
2	Ocean crust is 0 million years old at the ridge and 120 million years old 3,000 km away (2.5 cm per year).	Age increases with distance from the ridge.
2	North American crust is about 2.7 billion years old at its core and about 400 million years old in the Appalachians.	Continents have old cores and younger edges.
3	Meteorites are about 4.56 billion years old; the oldest Earth rock is about 4.0 billion.	Earth's early record is missing.
4	Ocean crust is about 3.0 g/cm³; continental crust is about 2.7 g/cm³.	The denser ocean plate sinks; continents survive.
5	Kaua‘i moved about 10 cm per year; GPS says 7 to 8 cm per year.	Two independent methods agree.
6	The oldest ocean floor is only about 200 million years old.	Ocean crust is recycled.

My claim after Day 6:

Plate tectonics explains the ages very well: ocean crust is made at ridges and recycled at trenches, while less-dense continents survive and grow.

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Frayer card sheet (homework)

For each word write a definition in your own words, draw a quick picture, and give an example. Row 1 is done for you. Finish by Day 7.

Word	Definition in my own words	Picture	Example
lithosphere (plate)	A rigid slab of Earth's outer layer that moves.		The Pacific plate
plate boundary (divergent, convergent, transform)	Where two plates meet: they move apart, together, or past each other.	sketch: three arrow pairs	San Andreas Fault
mid-ocean ridge	Underwater mountain chain where new ocean crust forms.	sketch: ridge with arrows	Mid-Atlantic Ridge
subduction	One plate sinks beneath another.	sketch: slab diving down	Nazca plate under South America
trench	A deep groove where a plate sinks.	sketch: V-shaped groove	Mariana Trench
magnetic reversal	Earth's magnetic field flips direction; the crust records it as stripes.	sketch: mirror stripes	Stripes on each side of a ridge
craton (a continent's ancient core)	The ancient core of a continent.	sketch: bullseye of ages	Canadian Shield
impact crater	A bowl-shaped dent made when a rock hits a surface.	sketch: crater	Moon highlands

UNIT 3.1 · DAY 1 · GLUE IN · REFERENCE

Hook data: ocean floor and continent ages

Place on the ocean floor	Age of the rock
Right at the Mid-Atlantic Ridge	0 (forming now)
Middle of the Atlantic, about 2,000 km from the ridge	about 80 million years
Atlantic seafloor next to the east coast of North America	about 180 million years
Right at the East Pacific Rise	0 (forming now)
Western Pacific, east of Japan (the oldest on Earth)	about 180 million years

Continental material	Where	Age
Oldest mineral (a zircon crystal)	Jack Hills, Western Australia	about 4.4 billion years
Oldest known rock (Acasta Gneiss)	Northwest Territories, Canada	about 4.0 billion years
Very old rock (Isua belt)	Greenland	about 3.8 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Site list for the plotted map

Plot each site on your world map: a red dot for a volcano, a blue X for an earthquake. Each partner plots five. Check Plotted when done. Mount St. Helens is done for you.

Site	Type	Latitude	Longitude	Plotted
Mount St. Helens, Washington, USA	volcano	46.2° N	122.2° W	✓
Mount Fuji, Japan	volcano	35.4° N	138.7° E	✓
Popocatépetl, Mexico	volcano	19.0° N	98.6° W	✓
Hekla, Iceland	volcano	64.0° N	19.7° W	✓
Erta Ale, Ethiopia	volcano	13.6° N	40.7° E	✓
Tohoku, Japan (2011, magnitude 9.1)	earthquake	38.3° N	142.4° E	✓
Maule, Chile (2010, magnitude 8.8)	earthquake	35.9° S	72.7° W	✓
Haiti (2010, magnitude 7.0)	earthquake	18.4° N	72.5° W	✓
Turkey (2023, magnitude 7.8)	earthquake	37.2° N	37.0° E	✓
Kīlauea, Hawai'i, USA	volcano	19.4° N	155.3° W	✓

The points make lines. They match the edges of the plates.

Kīlauea is a hot spot inside a plate: circle it as the one that does not fit.

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT

Plate boundary table

Fill in the table after you make each clay model. Divergent is done for you.

Boundary	What the plates did	What formed	Real example
Divergent	Moved apart	New crust at a mid-ocean ridge	Mid-Atlantic Ridge
Convergent	Pushed together; the ocean slab slid under	A trench and volcanoes (subduction)	Pacific coast of South America
Transform	Slid past each other	A broken line; earthquakes	San Andreas Fault

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE

Seafloor ages and color key

Distance from ridge (km)	Crust age (million years)
0	0
500	20
1,000	40
1,500	60
2,000	80
2,500	100
3,000	120

Age (million years)	Color
0	red
20	orange
40	yellow
60	green
80	blue
100	purple
120	dark purple

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT

Seafloor strip

Color each box using the key on D2-A. The ridge is done for you.

3,000 km	2,500	2,000	1,500	1,000	500	RIDGE	500	1,000	1,500	2,000	2,500	3,000 km
dark purple	purple	blue	green	yellow	orange	red	orange	yellow	green	blue	purple	dark purple

As you move away from the ridge, the crust gets **older** .

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT

Graph grid: age vs. distance

Mark a dot in the box where the distance column meets the age row. Connect the dots. The first dot (0 km, 0 years) is done for you.

Age (million years)	0 km	500	1,000	1,500	2,000	2,500	3,000 km
120							●
100						●	
80					●		
60				●			
40			●				
20		●					
0	●						

Rate = distance ÷ time = **25** km per million years = **2.5** cm per year. Age at 3,500 km: **140** million years.

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE AND WRITE ON IT

North America crust ages and strip

Color the oldest region dark brown and the youngest light yellow. Choose three colors in between. The Canadian Shield is done for you.

Region of North America	Direction and distance from the central core (km)	Age of the crust
Pacific Coast Ranges	west, about 2,800	about 50 million years
Rocky Mountain region	west, about 1,800	about 150 million years
Canadian Shield (the central core)	0	about 2.7 billion years
Grenville Province (Great Lakes to Quebec)	east, about 1,200	about 1.1 billion years
Appalachian Mountains	east, about 2,000	about 400 million years

Pacific Coast Ranges (west)	Rocky Mountains	Canadian Shield (core)	Grenville Province	Appalachians (east)
light yellow	yellow	dark brown	brown	tan

UNIT 3.1 · DAY 3 · GLUE IN · WRITE ON IT

Crater counts

Count the craters at each time step in the simulation. Row 1 is done for you. Then answer the questions.

Time step	Moon craters	Earth craters
1 billion years	6	5
2 billion years	12	4
3 billion years	18	6
4 billion years	24	5

Which surface looks older? How can you tell?

The Moon: it has many craters that stay.

Why does Earth show so few craters?

Erosion, volcanoes, and moving plates erase them.

A surface has a lot of craters. Is it old or young? Why?

Old: craters pile up over time.

UNIT 3.1 · DAY 3 · GLUE IN · REFERENCE AND WRITE ON IT

Timeline table and old materials

Scale: 1 cm = 100 million years. Position = (4,500 − age in millions of years) ÷ 100. Two rows are done for you.

Event	Age (millions of years ago)	Position on the 45 cm strip (cm)
Earth forms (left end)	4,500	0
Oldest mineral on Earth (zircon)	4,400	1
Oldest known rock	4,000	5
Oldest fossils	3,500	10
Oldest ocean floor	200	43
Today (right end)	0	45

Material	Where	Age
Meteorite fragments (Canyon Diablo)	Arizona, USA	about 4.56 billion years
Moon rocks from the lunar highlands	Brought back by Apollo	up to about 4.4 billion years
Oldest Earth mineral (zircon)	Jack Hills, Australia	about 4.4 billion years
Oldest known Earth rock (Acasta Gneiss)	Canada	about 4.0 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years
Oldest ocean floor	Western Pacific and Atlantic margins	about 180–200 million years

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

Guided notes

Fill in the blanks as your teacher goes. Blank 1 is done for you.

1. Old ocean crust is made at a **mid-ocean ridge** and moves away about 2.5 cm a year (25 km per million years).
2. A piece of crust travels 3,000 km. Time = 3,000 ÷ 25 = 120 million years. In 200 million years it travels 200 × 25 = 5,000 km, so it reaches a trench and sinks. No ocean floor is older than about 200 million years.
3. Ocean crust has a density of about **3.0 g/cm³**. Continental crust is about **2.7 g/cm³**. The ocean plate sinks (*subduction*).
4. Each continent has an ancient core. Volcanic arcs and pieces of crust are added to its edge over time, like rings on a tree. The edge is younger than the core.

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

Density table

Water has a density of 1.0 g/cm³. Density = mass ÷ volume.

Block	Mass (g)	Volume (cm³)	Density (g/cm³) = mass ÷ volume	Sinks or floats?
Wood block (stand-in for a continent)	40	64	0.6	floats
Clay block (stand-in for an old, cold ocean plate)	128	64	2.0	sinks

Which block is denser? clay What does the denser one do when they meet? dives under

Sample values; real numbers depend on your blocks.

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT

SketchNotes checklist

Check each box when it is in your SketchNotes. Turn in a photo on Canvas: Lesson 4: SketchNotes.

	Half page of notes		Cross-section drawing
☑	Where ocean crust is made	☑	Ridge and plate moving away
☑	How it moves and sinks	☑	Trench and volcanic arc
☑	How continents grow from a core	☑	Continent with an ancient core and a younger edge
☑	Density numbers	☑	Density labels and a carbon arrow

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE

Magnetic stripe bands

Color both strips with this table (2 cm = 1 million years). Start at the end you will push through first.

Band (from the end you push first)	Color	Length	Age it represents
1	Reversed (blue)	3.4 cm	about 3.6–5.3 million years
2	Normal (red)	2.0 cm	about 2.6–3.6 million years
3	Reversed (blue)	3.6 cm	about 0.8–2.6 million years
4	Normal (red)	1.6 cm	0–0.8 million years (youngest)

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE AND WRITE ON IT

Hawaii and GPS

1 km per million years = 0.1 cm per year. GPS today: about 7 to 8 cm per year. The first rate is worked for you.

Volcano or island	Distance northwest of Kīlauea (km)	Age of its oldest rock (million years)
Kīlauea (Hawai'i Island, active now)	0	0
Kaua'i	about 500	about 5
Midway Atoll	about 2,500	about 28

Kīlauea to Kaua'i: 500 ÷ 5 = **100** km per million years = **10** cm per year (done for you).

Kīlauea to Midway: 2,500 ÷ 28 = about 90 km per million years = about 9 cm per year.

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE

Fossil table

Organism	What it was	Found in
Mesosaurus	freshwater reptile	South America and southern Africa
Glossopteris	seed fern (heavy seeds)	South America, Africa, India, Australia, Antarctica
Lystrosaurus	land animal	Africa, India, Antarctica

UNIT 3.1 · DAY 5 · GLUE IN · WRITE ON IT **Evaluate the evidence**

Rate strength from 1 (weak) to 4 (very strong). Row 1 is started for you.

Line of evidence	Measured numbers?	Past or present motion?	Independent of the others?	Strength (1–4)
Rock ages across a ridge	Yes	Past	Yes	4
Magnetic stripes	Yes	Past	Yes	4
GPS and Hawaiian island ages	Yes	Present and past	Yes	4
Matching fossils and rocks	No	Past	Partly	2 to 3

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Three buckets**

Put each row of your Unit CER log in a bucket. One example is done for you.

The seafloor is young because…	The continents are old because…	The early record is missing because…
The oldest ocean floor is only about 200 million years old (done for you).		
Crust is 0 at the ridge and 120 million years at 3,000 km.	The Canadian Shield is about 2.7 billion years old.	Plate tectonics and erosion recycled it.
Ocean plates sink at trenches.	Continents are less dense, so they stay on top.	Scientists use meteorites and moon rocks.

UNIT 3.1 · DAY 6 · GLUE IN · REFERENCE **Part A sentence frames**

1. **Claim:** Plate tectonics explains the ages of Earth's crustal rocks (very well / partly / poorly) because _____.
2. **Ocean evidence:** The seafloor is youngest at the _____ and gets older with _____; the oldest ocean floor is only about _____ million years old.
3. **Continent evidence:** Continental crust as old as _____ exists, and in North America the crust gets _____ toward the edges.
4. **Reasoning:** Because ocean plates are _____ than continental crust, they _____ at subduction zones, while continents _____.
5. **Limit or counterclaim:** One thing this evidence cannot show is _____. OR: Someone might say the ocean floor is young because _____. This does not fit the evidence because _____.

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Peer feedback**

Trade papers. Give one star (something strong) and one step (one thing to improve).

Partner's name	One star	One step
Sam	You used the number 2.5 cm per year.	Say what the evidence cannot show.

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Evidence cards**

Cut the cards apart. Put them in order from oldest to newest on a paper strip and glue them down. Card 1 is placed for you.

Order	Evidence card	What it shows
1	Meteorites: about 4.56 billion years	The solar system formed about 4.56 billion years ago.
2	Moon highland rocks: up to about 4.4 billion years; heavily cratered	Solid crust existed early; the early solar system had many impacts.
3	Earth's oldest zircons: about 4.4 billion years	Solid crust and minerals had formed on Earth by then.
4	Oldest Earth rock (Acasta Gneiss): about 4.0 billion years	Rock has survived for 4 billion years in the continents.
5	Oldest fossils (stromatolites): about 3.5 billion years	Life existed by then.
6	Oldest ocean floor: about 200 million years	Ocean crust is recycled; the early record is not on the ocean floor.

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Part B story frames**

Use your ordered evidence cards. Frame 1 is done for you.

1. About **4.56** billion years ago, **the solar system formed**. The evidence is **meteorites**.
2. About **4.4** billion years ago, **solid crust and zircon minerals formed**. The evidence is **moon rocks and zircons**.
3. About **3.5** billion years ago, **life existed**. The evidence is **stromatolite fossils**.
4. Earth's earliest record is missing because **plate tectonics and erosion recycled it**.

UNIT 3.1 · DAY 7 · GLUE IN · WRITE ON IT **Self-rating chart**

Rate each target from 1 (I need help) to 4 (I could teach it). Use your CFA scores to check.

Target	My rating (1–4)	My CFA score	What I will do today
ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.	(personal)	(personal)	(personal)
ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.	(personal)	(personal)	(personal)
ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.	(personal)	(personal)	(personal)
ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.	(personal)	(personal)	(personal)
ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.	(personal)	(personal)	(personal)

Day 1: Where Did the Old Ocean Floor Go?

- Sites and boundary type: Mount St. Helens: convergent (Juan de Fuca plate subducting under North America); Mount Fuji: convergent (subduction); Popocatépetl: convergent (Cocos plate subducting under North America); Hekla: divergent (Mid-Atlantic Ridge on land, with a hot spot); Erta Ale: divergent (Afar rift); Tohoku: convergent (Pacific plate subducting under Japan); Maule: convergent (Nazca plate subducting under South America); Haiti: transform (strike-slip fault); Turkey: transform (East Anatolian Fault); Kīlauea: NOT at a plate boundary: a hot spot in the middle of the Pacific plate.
- Pattern: sites form lines along plate boundaries; the Ring of Fire is the largest.

- Clay models: divergent — plates move apart, new crust forms at a mid-ocean ridge; convergent — the denser ocean slab slides under the continent, a trench forms, volcanoes rise; transform — plates slide past each other, the line breaks, earthquakes.
- Crust is made at mid-ocean ridges (divergent boundaries) and destroyed at trenches (subduction at convergent boundaries).
- Sample first claim: The ocean floor is young because it keeps getting renewed (accept any attempt; this is the baseline).

Day 2: Reading Rock Ages

- Prediction: the sample at the ridge is younger; the Brazil sample is older (up to about 120 million years old).
- Strip colors: red 0, orange 20, yellow 40, green 60, blue 80, purple 100, dark purple 120. The youngest color is at the ridge; the oldest is at the outer boxes; the left and right sides match.
- Pattern sentence: As you move away from the ridge, the crust gets older.
- Graph: a straight line through the origin. Slope 500 km per 20 million years = 25 km per million years = 2.5 cm per year (about the speed fingernails grow).
- 3,500 km from the ridge: 3,500 ÷ 25 = 140 million years.
- Continents: the oldest crust is in the central core (Canadian Shield, about 2.7 billion years); the crust gets younger moving outward both east and west. Like the ocean, age changes with distance from a center; unlike the ocean, the ages are billions of years and the pattern is not the same on both sides.

Day 3: Reading Deep Time and Early Earth

- Simulation counts: Moon 6, 12, 18, 24 craters at 1, 2, 3, and 4 billion years; Earth 5, 4, 6, 5. The Moon surface looks older (more craters, still piling up); Earth stays nearly clear because erosion, volcanism, and plate tectonics keep renewing the surface. A surface with many craters is old.
- Earth shows few craters because plate tectonics, erosion, weathering, and volcanism erase them; the Moon highlands are oldest (most craters) and are the better archive of the early solar system.
- Timeline positions: oldest rock (4,000) = 5 cm; oldest fossils (3,500) = 10 cm; oldest ocean floor (200) = 43 cm. All existing ocean floor is in the last 2 cm of the 45 cm strip.
- The oldest Earth rocks are younger than Earth because Earth's earliest rocks were destroyed or recycled by plate tectonics and erosion.
- Meteorites formed with the solar system and have changed little, so their ages (about 4.56 billion years) are the best estimate; the Moon is geologically quiet and kept its early rocks.
- Fossils that old are in continental rock; ocean floor of that age has been subducted.

Day 4: Why the Ocean Is Young and the Continents Old

- Guided notes: ridge; 3,000 ÷ 25 = 120 million years; 200 × 25 = 5,000 km; about 200 million years; the ocean (denser) plate sinks; the edge is younger than the core.
- Lab: the clay block is denser (greater than 1 g/cm^3) and sinks; the wood block is less dense and floats. The ocean plate is like the clay, the continent is like the wood. When they meet, the denser one dives under. Water stands in for the mantle.
- Two continental plates collide: both are buoyant, so neither subducts; the crust crumples upward into mountains (such as the Himalaya).
- Continents survive because they are less dense; the ocean floor is recycled at trenches, so its age rarely exceeds about 200 million years.
- Edge crust is younger because volcanic arcs and pieces of crust were added to the edge of the continent later.
- Ocean crust: made at a ridge, moves, is destroyed in under 200 million years. Continental crust: ancient core, grows outward, survives for billions of years.
- Carbon is not lost: plankton shells carry carbon down; some returns to the air as CO_2 from volcanoes; the rest stays in the mantle for a long time.

Day 5: Evidence for Moving Crust

- Stripes are mirror images because crust forms at the ridge, records the magnetic field, then splits and moves apart both ways.

- Hawaii: Kauaʻi 500 km ÷ 5 million years = 100 km per million years = 10 cm per year. Midway 2,500 ÷ 28 ≈ 90 km per million years ≈ 9 cm per year. Similar to GPS (7 to 8 cm per year).
- Mesosaurus (freshwater) and Glossopteris (heavy seeds) could not cross oceans, so the lands were once joined. Fossils cannot give the rate or show present motion.
- Evaluation table (sample): rock ages — quantitative, past, independent, 4; stripes — quantitative, past, independent, 4; GPS and Hawaii — measured, present (GPS) and past (island ages), independent, 4; fossils and rocks — qualitative, past, partly independent, 2 to 3.
- Current motion: GPS. Past motion: rock ages and stripes agree, plus fossils.
- Two independent methods agree on a rate, which is hard to explain by chance, so the conclusion is more trustworthy.
- A weakening finding: crust at the ridge being old and getting younger away from it.

Day 6: Evaluate the Theory, Tell Earth's Story

- Sample strong claim: Plate tectonics explains the ages of Earth's crustal rocks very well because ocean crust is made at ridges and recycled at subduction zones, while less-dense continental crust survives and grows outward.
- Evidence examples: crust age increases from 0 to 120 million years across 3,000 km (2.5 cm per year); the oldest ocean floor is about 200 million years; continental minerals are up to about 4.4 billion years; North American crust is about 2.7 billion years at its core and about 400 million years in the Appalachians.
- Reasoning: dense ocean plates sink at trenches while buoyant continents stay at the surface and grow at their edges.
- Limit: no ocean floor older than about 200 million years remains, so early Earth must be studied in continental rock and in meteorites. Counterclaim: the ocean floor erodes away — sediment is thicker on older crust, so old crust is not worn away; it is subducted.
- Part B card order (oldest to newest): meteorites 4.56 billion; moon highlands up to 4.4 billion; zircons 4.4 billion; Acasta Gneiss 4.0 billion; stromatolites 3.5 billion; oldest ocean floor 200 million. The early record is missing because plate tectonics and erosion recycled it, so scientists use meteorites, the Moon, and other planets.

Day 7: Review Day

- Self-rating chart is personal. Compare it with CFA scores and discuss any mismatch.
- Review answers are in the teacher key (HS-ESS1-5_teacher_key.md in the repo).
- Enrichment answers: see the notes on the Enrichment assignment; for Option 1, 1 km ÷ 2.5 cm per year = 40,000 years; for Option 3, 2 cm/yr × 1 million years = 20 km, and 100 cm in 50 years; for Option 4, Mars's crust is older because it is not recycled.